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Problem 1

(a) Find the Lorentz transformations of charge density ρ and current density $\vec{j}$ (use O' moving with v along the x axis).

(b) Show that, under Lorentz transformations, the total charge

$$Q \equiv \int j^0(0, \vec{x}) dV = \int j^0(x) \delta(x^0) d^4x$$

is invariant; that is, $Q = Q'$, where

$$Q' = \int j^{0'}(x') \delta(x^{0'}) d^4x'.$$

Hint: Write $j^0(x) \delta(x^0) = j^\nu(x) \partial_\nu \theta(x^0)$, where $\theta(x^0)$ is the Heaviside function. You also will need $\Lambda^0_0 \geq 1$. Show that $Q - Q' = 0$. First show that

$$Q' = \int j^\mu(x) \partial_\mu \theta(x^{0'}) d^4x.$$

Then look at $Q' - Q$ and show that it vanishes if $\Lambda^0_0 \geq 1$.

θx^0 Heaviside function is just a unit step starting at 0.

A

We define $J^\mu = (c\rho, J_x, J_y, J_z)$

and

$$\Lambda = \begin{bmatrix} \gamma & -\gamma B & 0 & 0 \\ -\gamma B & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $B \equiv \frac{v}{c}$

charge density should be the same independent of coordinate frame to maintain the same physics, so

$\rho' = \rho$, and $\frac{j^0}{c} \equiv \rho$

to current density, we have

$$\begin{aligned} j^{\mu'} &= \Lambda^{\mu'}_\nu j^\nu \\ &= \Lambda^{\mu'}_\nu \rho u^\nu, \end{aligned}$$

where $u^\nu = (c\gamma, \dot{x}\gamma, \dot{y}\gamma, \dot{z}\gamma)$

We know how Λ transform velocity,

$$\begin{aligned} &\Lambda^{\mu'}_0 \rho u^0 + \Lambda^{\mu'}_i \rho u^i \\ &= \Lambda^{\mu'}_0 \rho \gamma v c + \Lambda^{\mu'}_j \rho \gamma u^j \end{aligned}$$

Λ takes $u^{0'} = \gamma(u^0 - Bu^1)$, $u^{1'} = \gamma(u^1 - Bu^0)$ and $u^{2'} = u^2$, $u^{3'} = u^3$.

Therefore, we have

$$\begin{aligned} j^{\mu'} &= \left(\rho \left(\gamma \left(u^0 - \frac{V}{c} u^1 \right) \right), \rho \left(\gamma \left(u^1 - \frac{V}{c} u^0 \right) \right), j^2, j^3 \right) \\ &= \left(\rho \gamma^2 \left(c - \frac{v}{c} \dot{x} \right), \rho \gamma^2 (\dot{x} - V), j^2, j^3 \right) \end{aligned}$$

dead work

$$j^{0'} = \underbrace{\gamma^2 \rho c}_0 + \rho \gamma^2 \left(\underbrace{c - \frac{v}{c} \dot{x}}_1 + \underbrace{\dot{x} - v}_2 + \dot{y} + \dot{z} \right)$$

B

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$j^0 = \rho c$, not just ρ .

Using the hint

$$Q = \int j^\nu \partial_\nu \theta(x^0) d^4x$$

$$Q' = \int j^{\nu'}(x') \partial_{\nu'} \theta(x^{0'}) d^4x'$$

$$Q' = Q$$

$$Q - Q' = \int j^\nu \partial_\nu \theta(x^0) d^4x - \int j^{\nu'} \partial_{\nu'} \theta(x^{0'}) d^4x'$$

We could drop the x' and $j^{\nu'}$ and d^4x' because

$$d^4x' = |\det(\Lambda)| d^4x \equiv d^4x$$

and because

$$T^{\mu'}_{\alpha', \nu'} = \Lambda^{\mu'}_\mu \Lambda^\alpha_{\alpha'} \Lambda^\nu_{\nu'} T^\mu_{\alpha, \nu}$$

and because summing the same index should end with a rank 0,0 tensor (which should be conserved in coordinate shifts).

We therefore can factor out

$$Q - Q' = \int (j^\nu(x) \partial_\nu) (\theta(x^0) - \theta(x^{0'})) d^4x$$

We can integrate by parts

$$= \int_{-\infty}^{\infty} \left(\frac{\partial}{\partial x_\mu} j^\nu(x) (\theta(x^0) - \theta(x^{0'})) \right) - \int_{-\infty}^{\infty} \frac{\partial j^\nu}{\partial x_\mu} (\theta(x^0) - \theta(x^{0'})) d^4x \xrightarrow{0}$$

The right is zero because conservation of charge.

We can just look at the argument of the first integral now.

$$\theta(x^0) - \theta(x^{0'}) \Big|_{-\infty}^{\infty}$$

$$x^{0'} = \Lambda^{0'}_0 x^0$$

The heaviside function is defined to be 1, so at infinity both of these are 1. We get 1-1 = 0 evaluated at infinity. At -infinity, each is zero. Therefore,

$$0 = 0$$

Work from before recitation

We can rewrite the Q' term

$$\begin{aligned} j^{\nu'}(x') \partial_{\nu'} \theta(x'^0) d^4 x' \\ = \Lambda_{\mu}^{\nu'} j^{\mu}(x) \partial_{\mu} \theta(x^0) d^4 x \end{aligned}$$

For $Q' = Q$, to be invariant, this quantity should be conserved for all $\Lambda_{\nu'}^{\nu}$.

That is,

$$j^{\nu} \partial_{\nu} \theta(x^0) d^4 x (1 - \Lambda_{\nu'}^{\nu}) = 0$$

This is trivially true for $x^0 < 0$. We can consider only $x \geq 0$.

$\int j^{\nu} \partial_{\nu} d^4 x (1 - \Lambda_{\nu'}^{\nu}) = 0$ must be true for all Λ where $x \geq 0$.

$$\int j^0 \frac{\partial(1 - \lambda_0^0)}{\partial x^0} dx + \int j^i \frac{\partial(1 - \lambda_i^{i'})}{\partial x^i} d^3 x$$

Problem 2

Define the 4-acceleration as $a^{\nu} = \frac{du^{\nu}}{d\tau}$.

(a) Show that $a^{\nu} u_{\nu} = 0$.

(b) Show that $a^{\nu} a_{\nu} \geq 0$.

(c) Show that

$$a^{\nu} = \gamma^4 \left(\frac{\vec{v}}{c} \cdot \vec{a}, \vec{a} + \frac{\vec{v}}{c} \times \left(\frac{\vec{v}}{c} \times \vec{a} \right) \right)$$

in the lab frame.

Hints: (a) Think of $u^{\nu} u_{\nu} = -c^2$; (b) consider the rest frame of the particle; (c) use the identity

$$\vec{w}_1 \times (\vec{w}_2 \times \vec{w}_3) = (\vec{w}_1 \cdot \vec{w}_3) \vec{w}_2 - (\vec{w}_1 \cdot \vec{w}_2) \vec{w}_3.$$

a

$$u^{\nu} u_{\nu} = c(-c) + 3 * 0(0)$$

$$a^{\nu} = \frac{du^{\nu}}{d\tau}$$

and

$$u_{\nu} = (-\gamma c, \gamma \dot{x}, \gamma \dot{y}, \gamma \dot{z})$$

$$a^{\nu} u_{\nu} = \left(-\gamma c \frac{d(\gamma c)}{d\tau}, \gamma \frac{d\dot{x}}{d\tau}, \gamma \frac{d\dot{y}}{d\tau}, \gamma \frac{d\dot{z}}{d\tau} \right)$$

Because we are in the rest frame, $\dot{x} = \dot{y} = \dot{z} = 0$

B

$$a^{\nu} a_{\nu} = -\frac{du^0}{d\tau}^2 + \frac{du^i}{d\tau}^2$$

We know that $u^0 = \gamma c$, and is constant (not changing with τ , so we are left with

$$a^{\nu} a_{\nu} = \frac{du^i}{d\tau}^2$$

For any argument u , because this is squared it will at minimum be 0.

C

(c) Show that

$$a^\nu = \gamma^4 \left(\frac{\vec{v}}{c} \cdot \vec{a}, \vec{a} + \frac{\vec{v}}{c} \times \left(\frac{\vec{v}}{c} \times \vec{a} \right) \right)$$

in the lab frame.

Hints: (a) Think of $u^\nu u_\nu = -c^2$; (b) consider the rest frame of the particle; (c) use the identity

$$\vec{w}_1 \times (\vec{w}_2 \times \vec{w}_3) = (\vec{w}_1 \cdot \vec{w}_3) \vec{w}_2 - (\vec{w}_1 \cdot \vec{w}_2) \vec{w}_3.$$

$$a^{\nu'} = \Lambda_{\nu'}^{\nu} \frac{du^\nu}{d\tau}$$

We know that Λ acts on ∂_ν as

Problem 3

Show that any solution to the equation

$$(y^2 - a^2) F(y) = 0, \quad a > 0,$$

can be written as

$$F(y) = f(y) \delta(y^2 - a^2)$$

for a convenient function $f(y)$.

Hints: Use the identities

$$f(x) \delta(x - a) = f(a) \delta(x - a), \quad \delta(x^2 - a^2) = \frac{1}{2|a|} (\delta(x - a) + \delta(x + a)).$$

The delta function makes any argument where $y^2 \neq a^2$ yield zero. In the case where $y^2 = a^2$, then the leading term is zero.

Lets call $\Gamma = (y^2 - a^2)$. The function is $f(y)\Gamma\delta(\Gamma)$

By definition, $\delta(\Gamma) = 0 \forall \Gamma \approx 0$.

Thus, we are just left with a spike with height $f(a)\Gamma$ where $\Gamma = 0$, aka the function has value 0. Therefore, for all y , this function is equal to zero.

We can rewrite this as

$$\frac{f(y)}{2|a|} (y^2 - a^2) (\delta(y - a) + \delta(y + a))$$

let C be the function $= \frac{f(y)}{2|a|}$

$$\begin{aligned} C[(y^2 - a^2)\delta(y - a) + (y^2 - a^2)\delta(y + a)] \\ = C[(a^2 - a^2)\delta(y - a) + (a^2 - a^2)\delta(y + a)] \\ = 0 \end{aligned}$$

Problem 4

Consider a generic spherically symmetric non-static four-current

$$j^0(t, \vec{x}) = \rho(t, x),$$
$$\vec{j}(t, \vec{x}) = \frac{\vec{x}}{x} j(t, x),$$

with $j^\mu(t, \vec{x}) = 0$ for $x > R, \forall t$.

Show that: the $\vec{E}$ and $\vec{B}$ in empty space for $x > R$ is static and given by

$$\vec{E} = Q \frac{\vec{x}}{x^3}, \quad \vec{B} = 0,$$

where

$$Q = \int \rho(t, x) dV.$$

Hint: The symmetry implies

$$\vec{E} = \vec{x} f(t, x), \quad \vec{B} = \vec{x} g(t, x).$$

We have the curl of $B = 0$ and the divergence of $B = 0$, so we have $\vec{B} = 0$.

For E, we just have

$$\Phi_E = \oint \oint \vec{E} \cdot d\vec{A} = 4\pi Q_{enc}$$

Because of spherical symmetry, $\vec{E}$ is always pointed normal to the sphere, which is parallel with $d\vec{A}$. Therefore, we have

$$|E| 4\pi r^2 = 4\pi Q_{enc}$$
$$|E| = \frac{Q_{enc}}{r^2}$$

We have the ansatz

$$|E| = \hat{r} f(t, r)$$
$$\frac{Q_{enc}}{r^2} = |r| \hat{r} f(t, r)$$
$$f(t, r) = \frac{Q_{enc}}{r^3}$$

This function does not have any time dependence, only r dependence.